```
Zimbabwe, step 2 (2026-11-02)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.279             -42584.1            -42573.4                 0
     2 20263017             5000           0.279             -42693.6            -42681.9                 0
     3 20263018             5000           0.281             -42536.4            -42528.3                 0
     4 20263019             5000           0.279             -42505.7            -42505.5                 0
                                      file
 mcmc_zimbabwe_step2_chain1_2026-10-31.Rdata
 mcmc_zimbabwe_step2_chain2_2026-10-31.Rdata
 mcmc_zimbabwe_step2_chain3_2026-11-01.Rdata
 mcmc_zimbabwe_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
age.65.to.79.mortality.intercept.multiplier.female               over.50.hiv.mortality.multiplier.2
                                             2.615                                            2.357
                         age.25.to.50.aging.factor                   hiv.specific.mortality.rates.2
                                             2.302                                            2.266
            age.40.to.49.transmission.multiplier.0            age.15.to.19.transmission.multiplier.1
                                             2.238                                            2.165
                                            trate.3            age.30.to.39.transmission.multiplier.2
                                             2.085                                            2.009
       over.80.mortality.intercept.multiplier.male            age.0.to.14.hiv.mortality.multiplier.1
                                             1.998                                            1.933
age.45.to.64.mortality.intercept.multiplier.female            age.20.to.29.transmission.multiplier.0
                                             1.891                                            1.885
           age.25.to.49.hiv.mortality.multiplier.1            age.0.to.14.hiv.mortality.multiplier.2
                                             1.800                                            1.717
             age.20.to.29.transmission.multiplier.1                male.hiv.mortality.multiplier.0
                                             1.716                                            1.714
         age.50.and.over.transmission.multiplier.3                 hiv.specific.mortality.rates.0
                                             1.686                                            1.597
             age.15.to.19.transmission.multiplier.2                male.hiv.mortality.multiplier.1
                                             1.575                                            1.556
             age.30.to.39.transmission.multiplier.0          age.15.to.24.hiv.mortality.multiplier.2
                                             1.551                                            1.535
                    male.hiv.mortality.multiplier.2                      male.awareness.multiplier
                                             1.511                                            1.486
           age.25.to.49.hiv.mortality.multiplier.0
                                             1.478

Merged simset:
 location simulations total.weight                                                file
 Zimbabwe        1000         1000 simset_zimbabwe_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 53 of 63 targets flagged (worst 30 shown; all in <location>_fit_check.csv)
     data.type         age    sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality       25-49 female              26                   0              x0.48      2025            x0.01
 hiv.mortality       25-49   both              26                   8              x0.59      2025            x0.00
 hiv.mortality       15-24   male              26                  31              x0.57      2025            x0.25
     awareness        0-14   both              11                   0          +19.4 pts      2025        +31.0 pts
     incidence 50 and over   male              26                  35              x0.61      2025            x0.97
 hiv.mortality       25-49   male              26                   4              x0.68      2025            x0.00
 hiv.mortality       10-19   male              26                  12              x0.69      2025            x0.33
     incidence 50 and over   both              26                   8              x0.74      2025            x1.32
     incidence       10-19   male              26                   4              x0.76      2025            x0.47
```

Zimbabwe step 2: likelihood

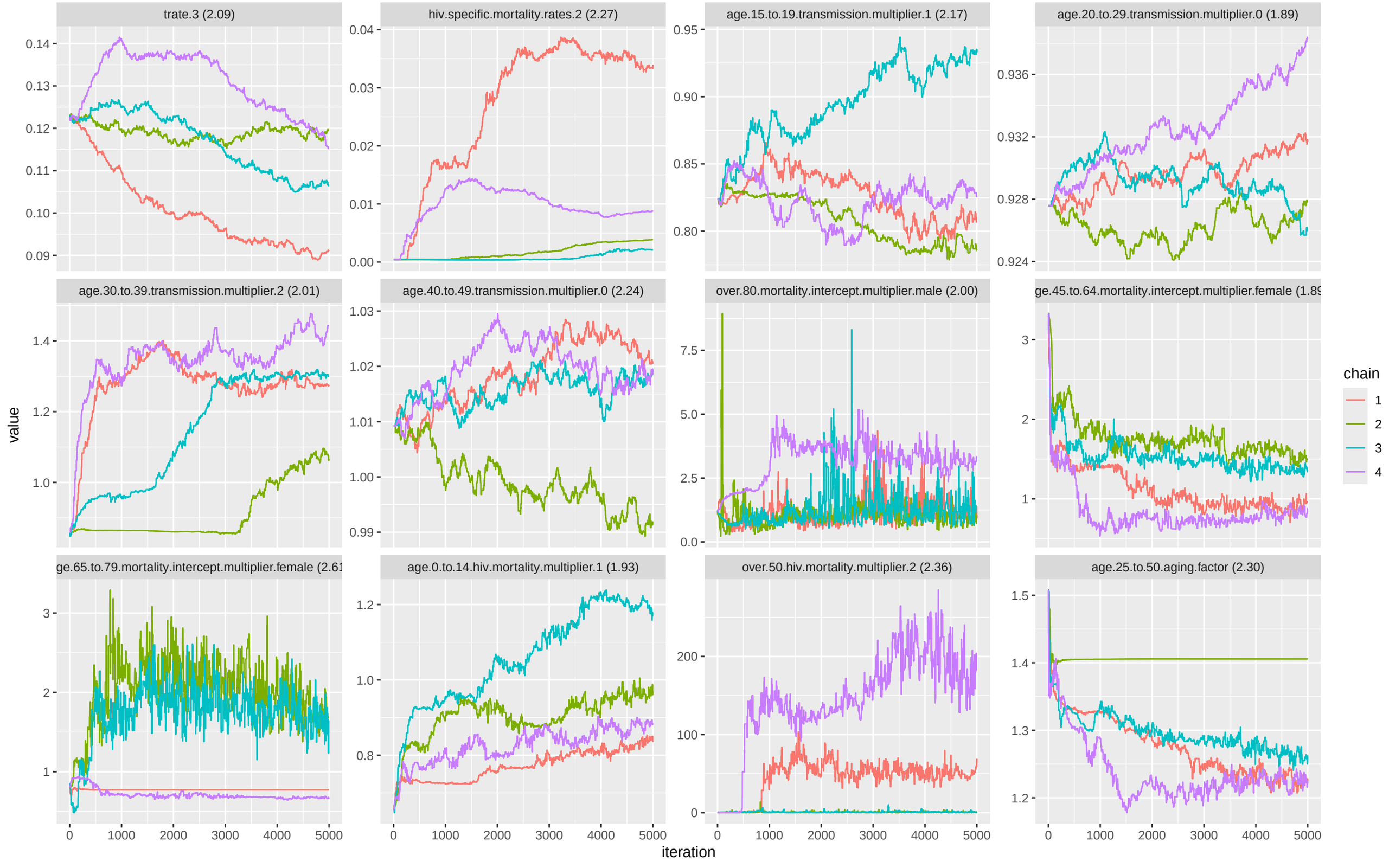
0
−10000
−20000
−30000
−40000
lue

0
25000
50000
75000
100000
iteration

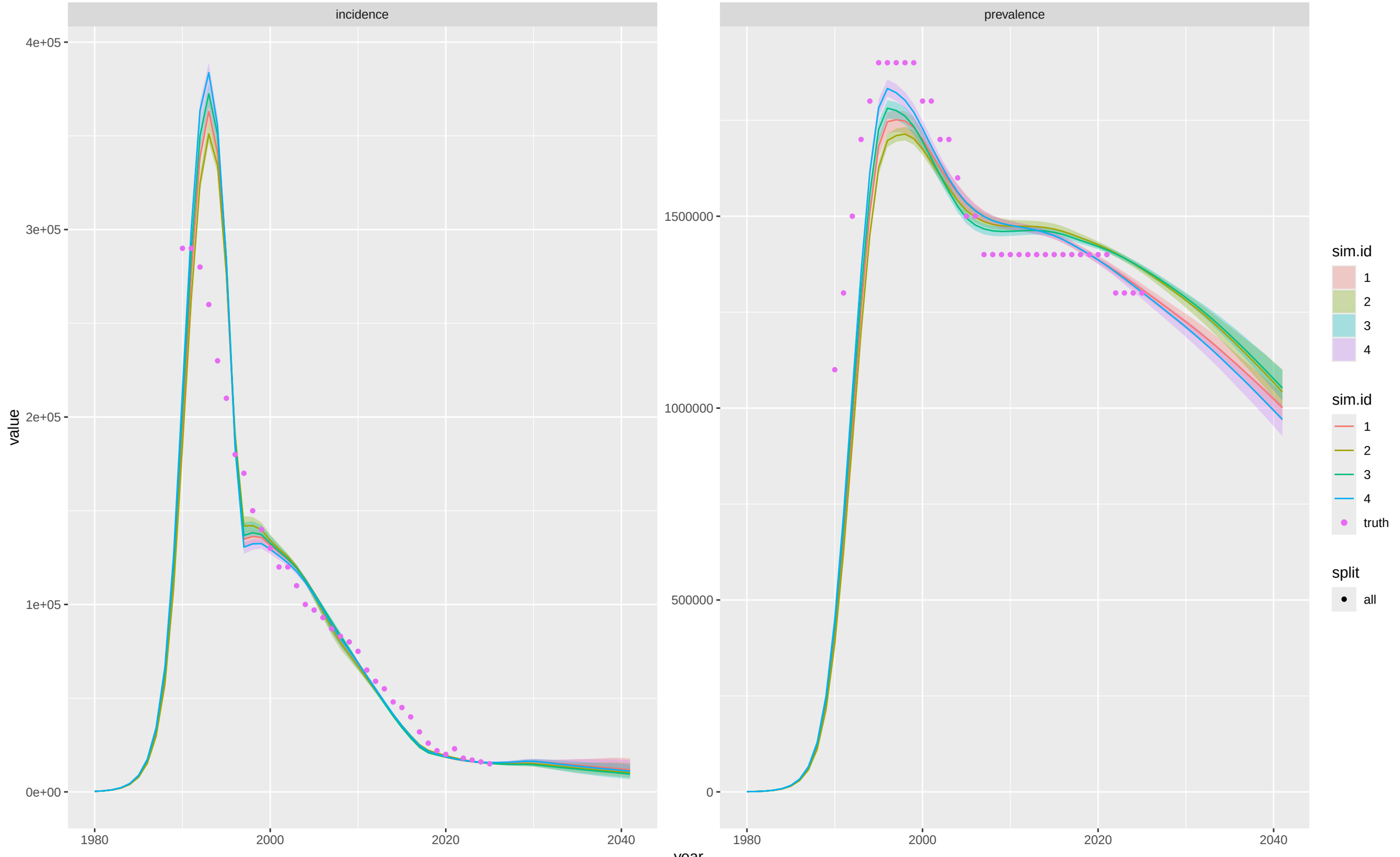
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

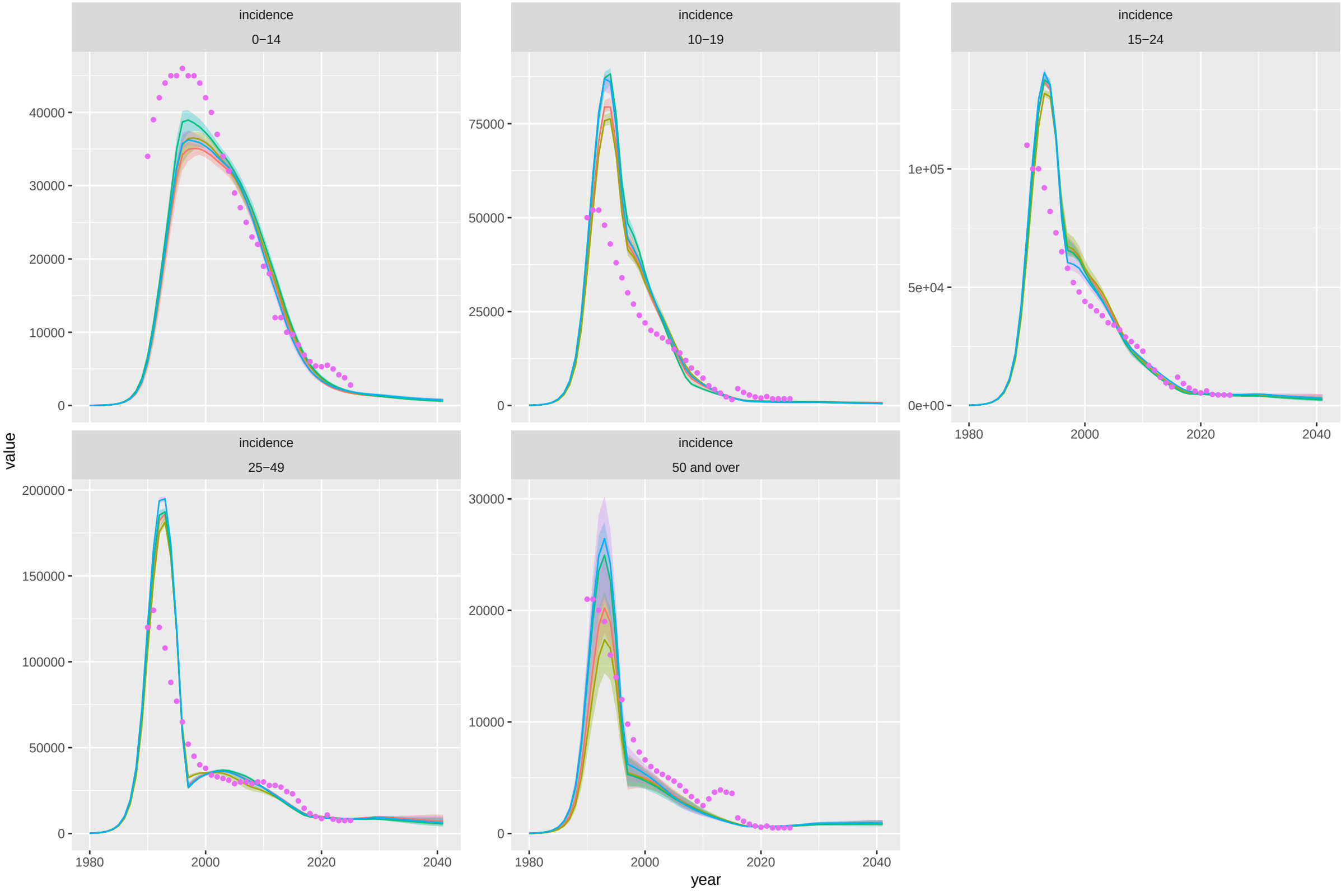
Zimbabwe step 2: trace



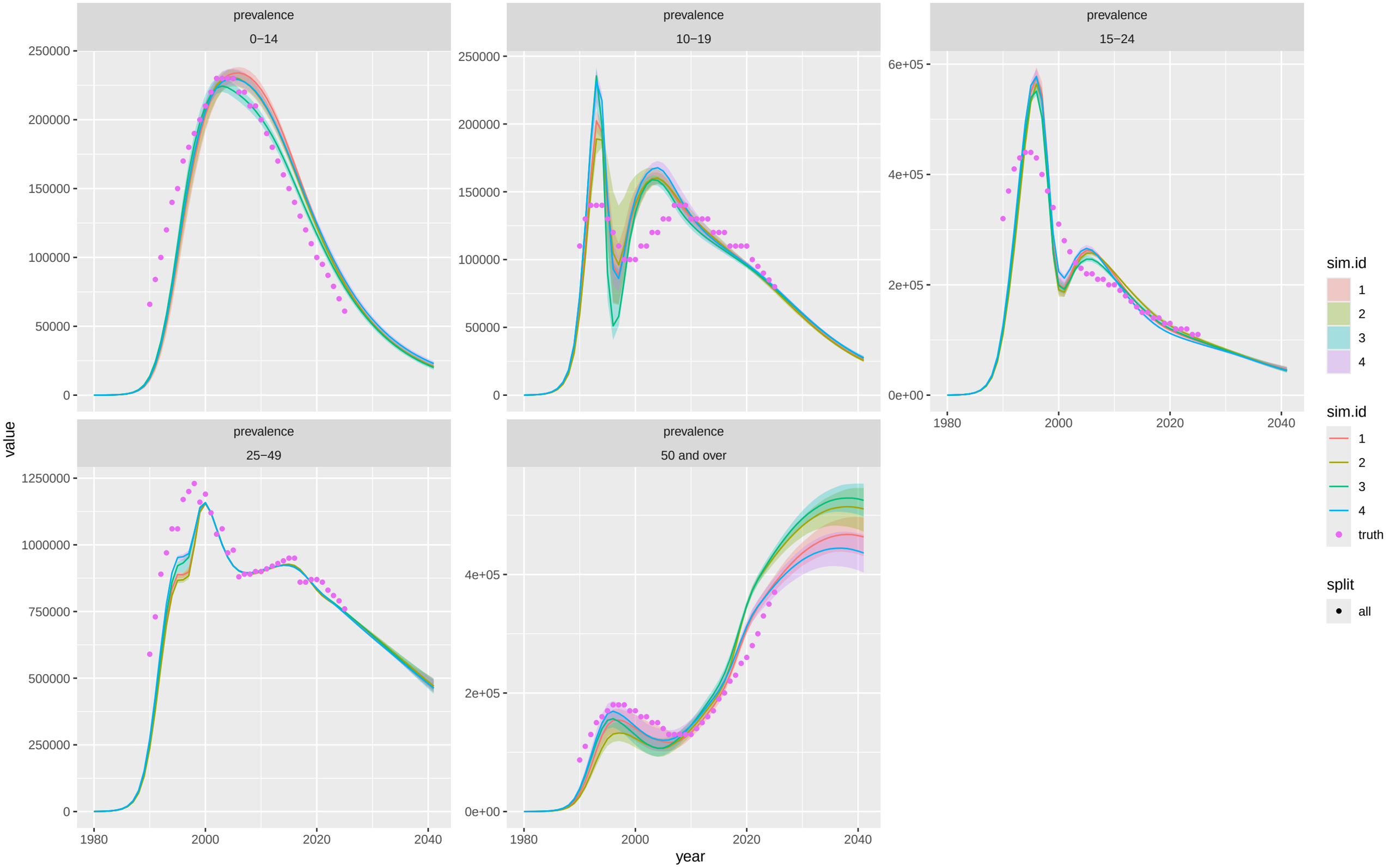
Zimbabwe step 2: fit (one color per chain)



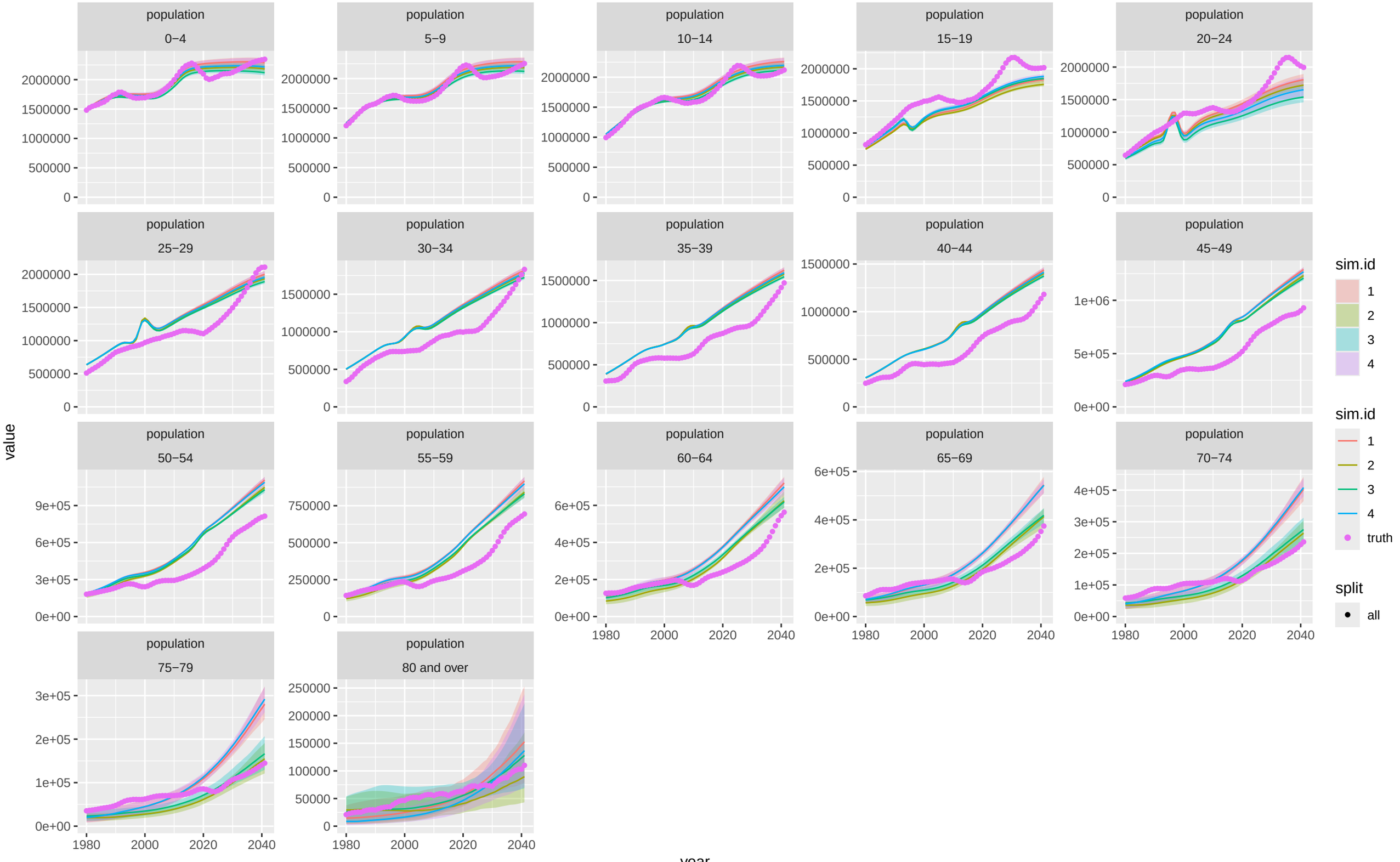
Zimbabwe step 2: incidence.by.age (one color per chain)



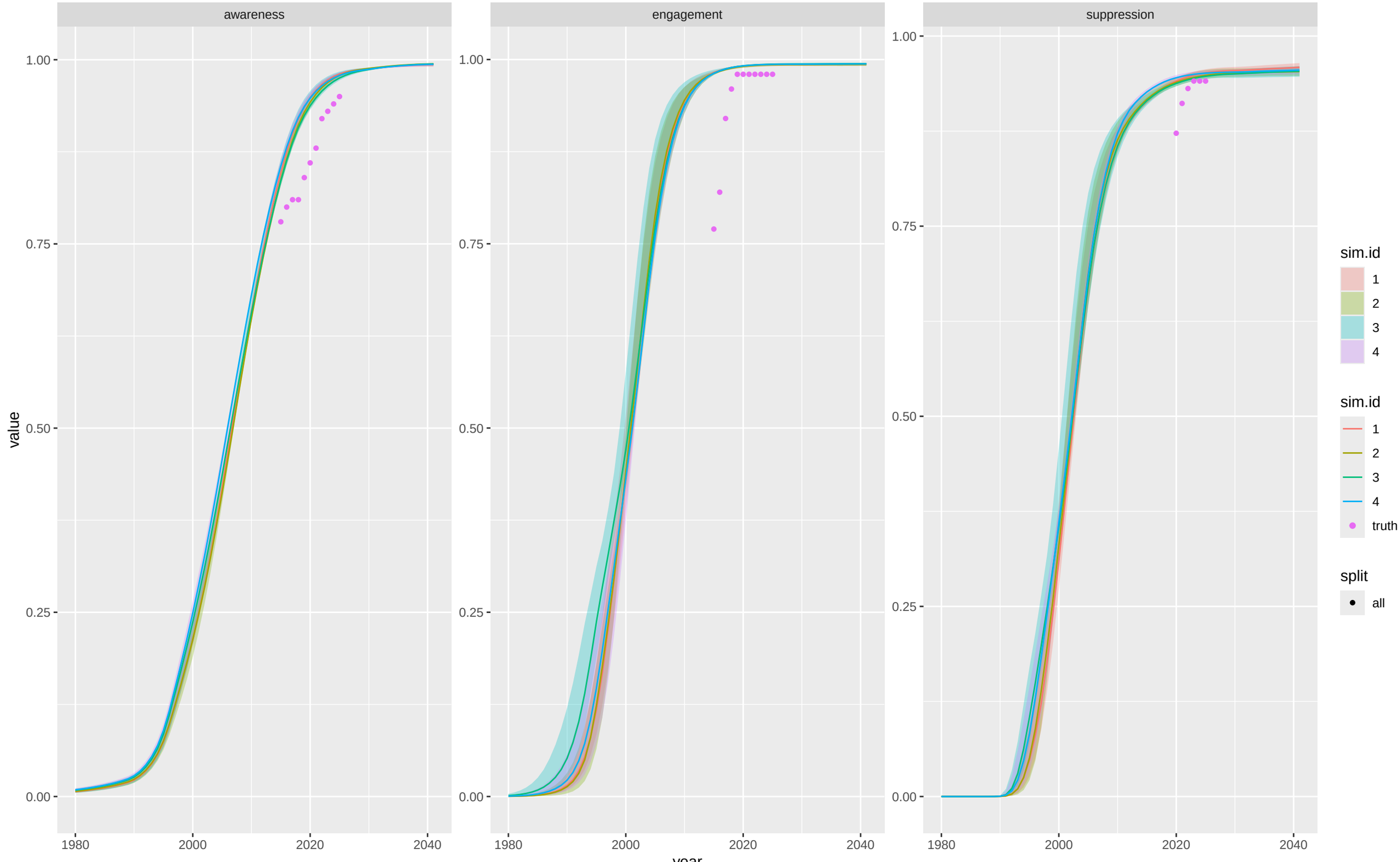
Zimbabwe step 2: prevalence.by.age (one color per chain)



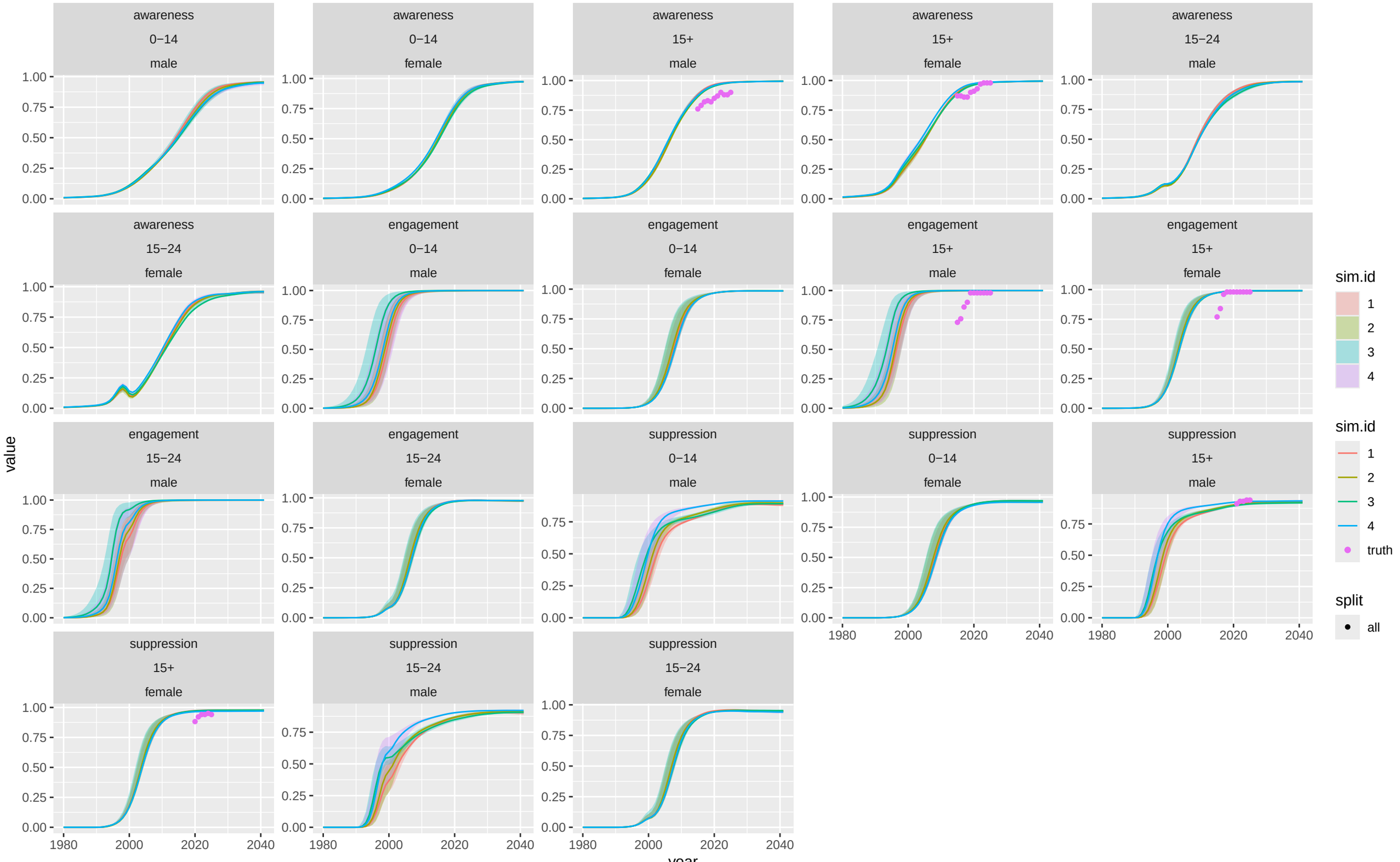
Zimbabwe step 2: population.by.age (one color per chain)



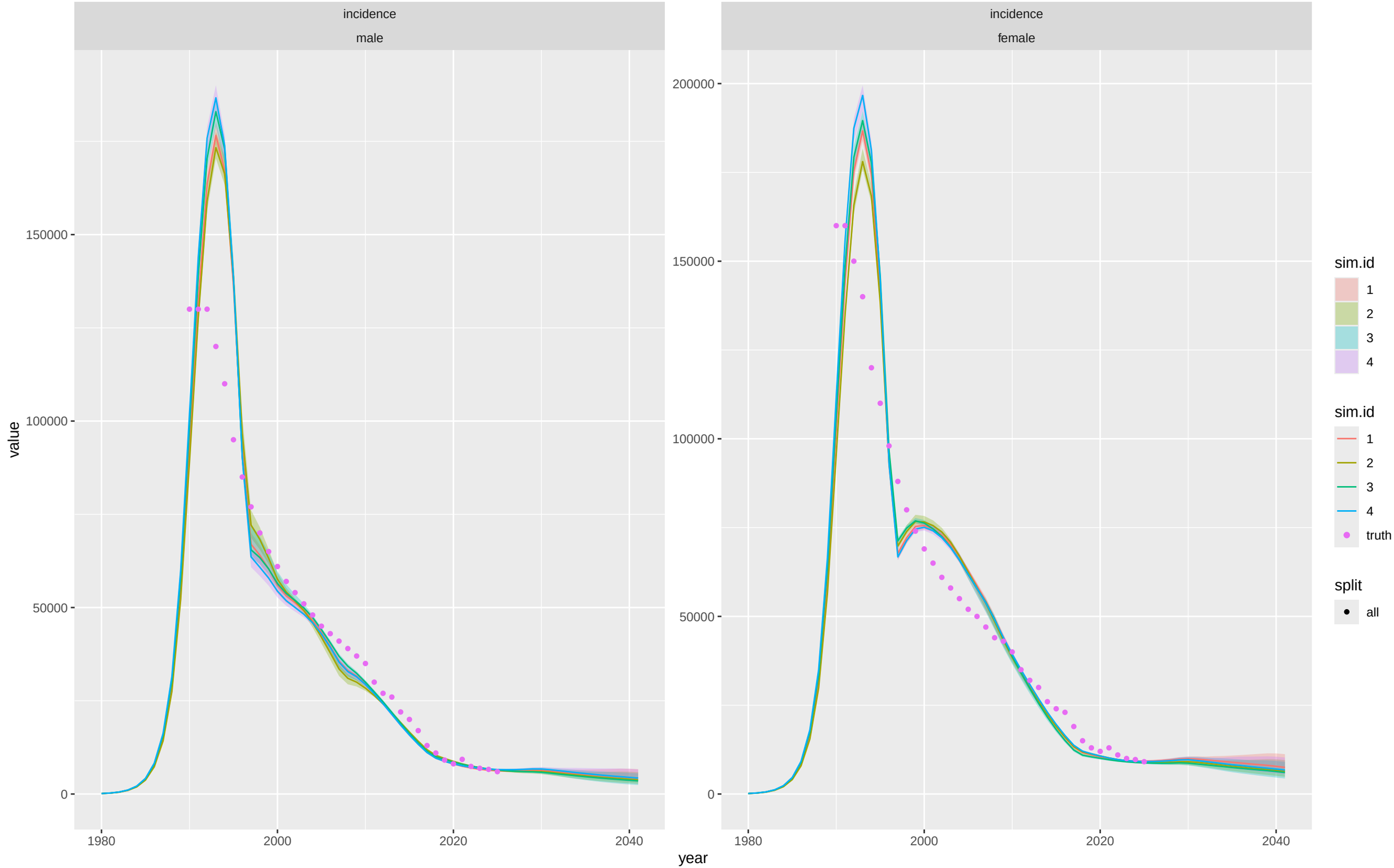
Zimbabwe step 2: cascade (one color per chain)



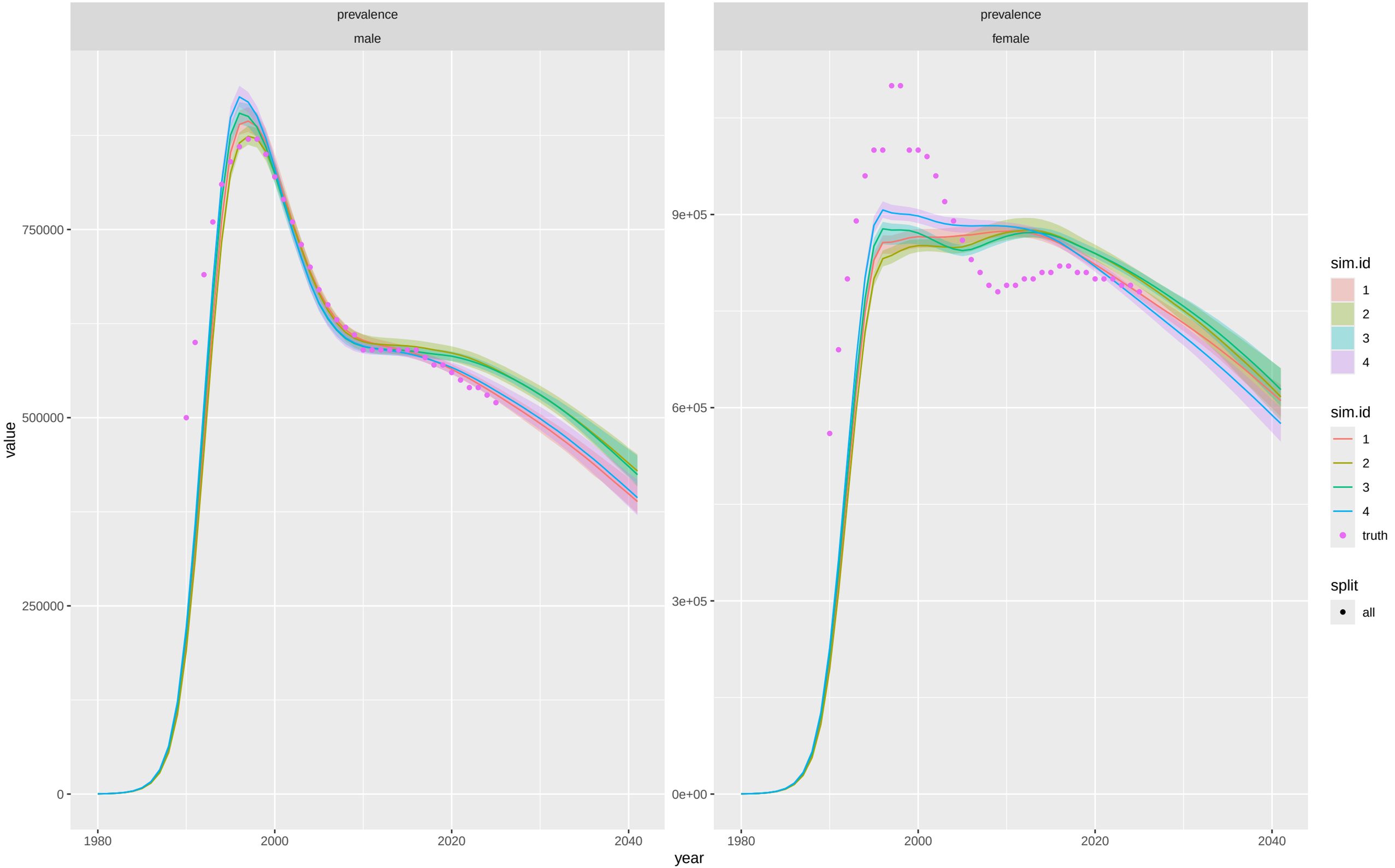
Zimbabwe step 2: cascade.by.age.sex (one color per chain)



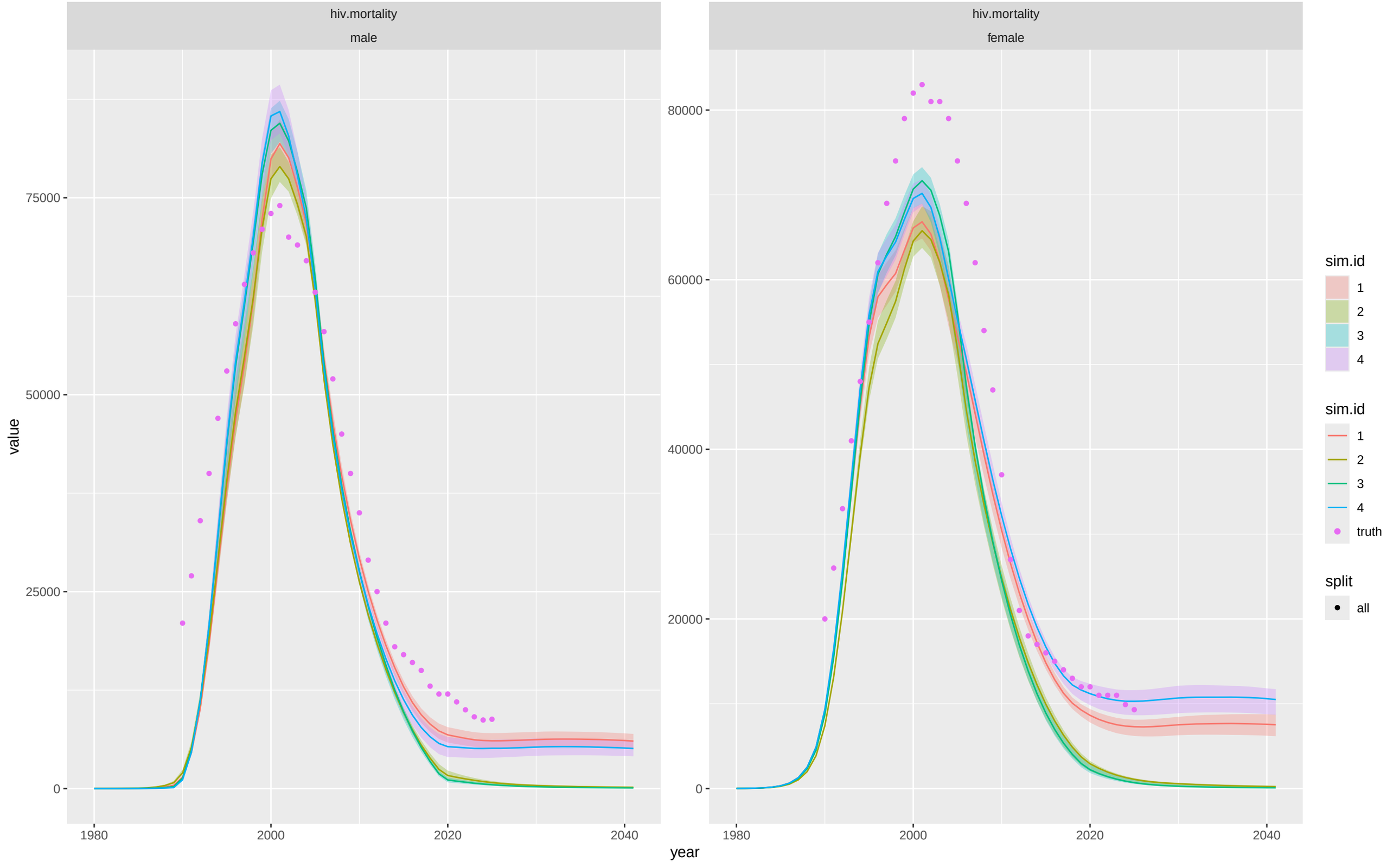
Zimbabwe step 2: incidence.by.sex (one color per chain)



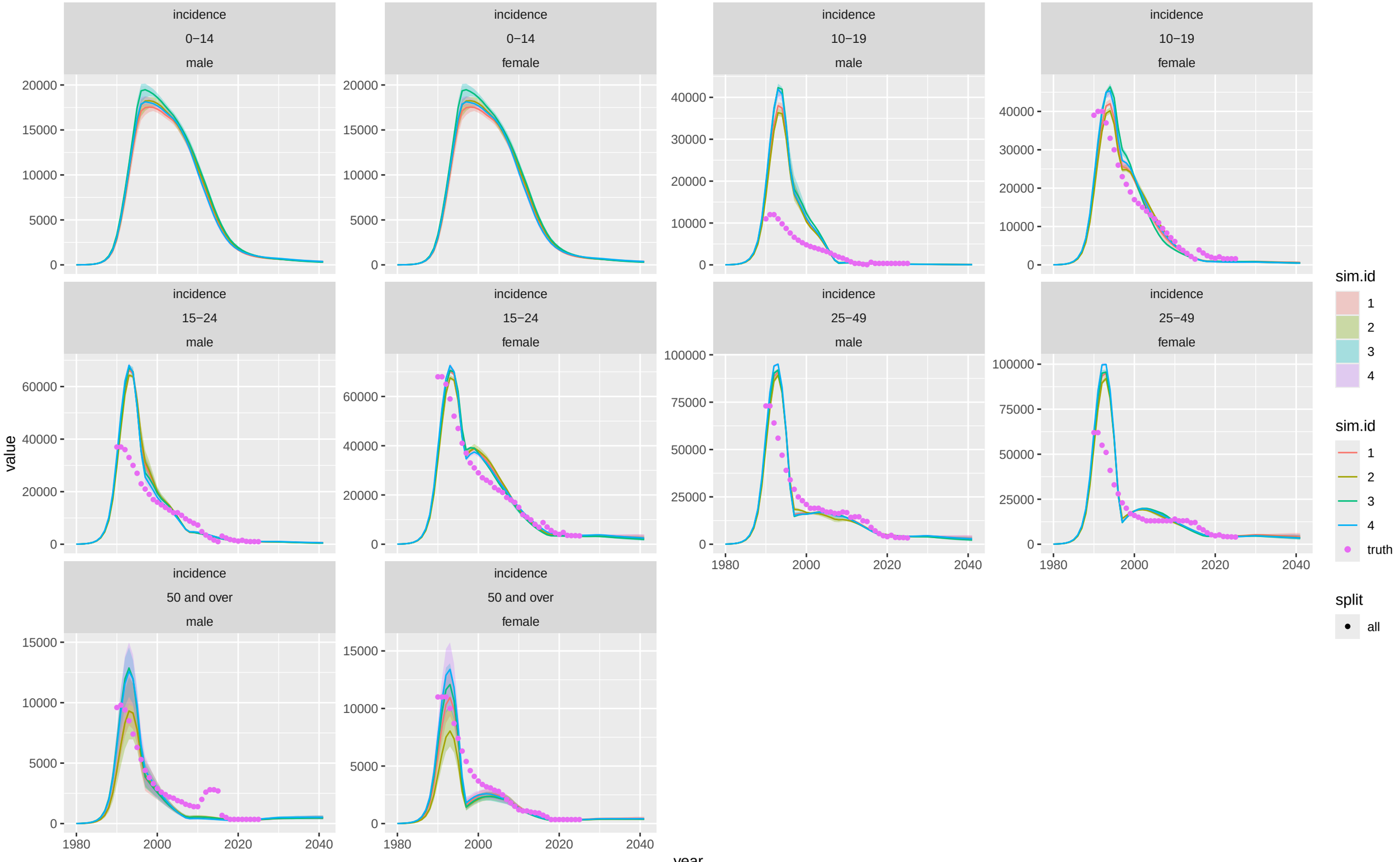
Zimbabwe step 2: prevalence.by.sex (one color per chain)



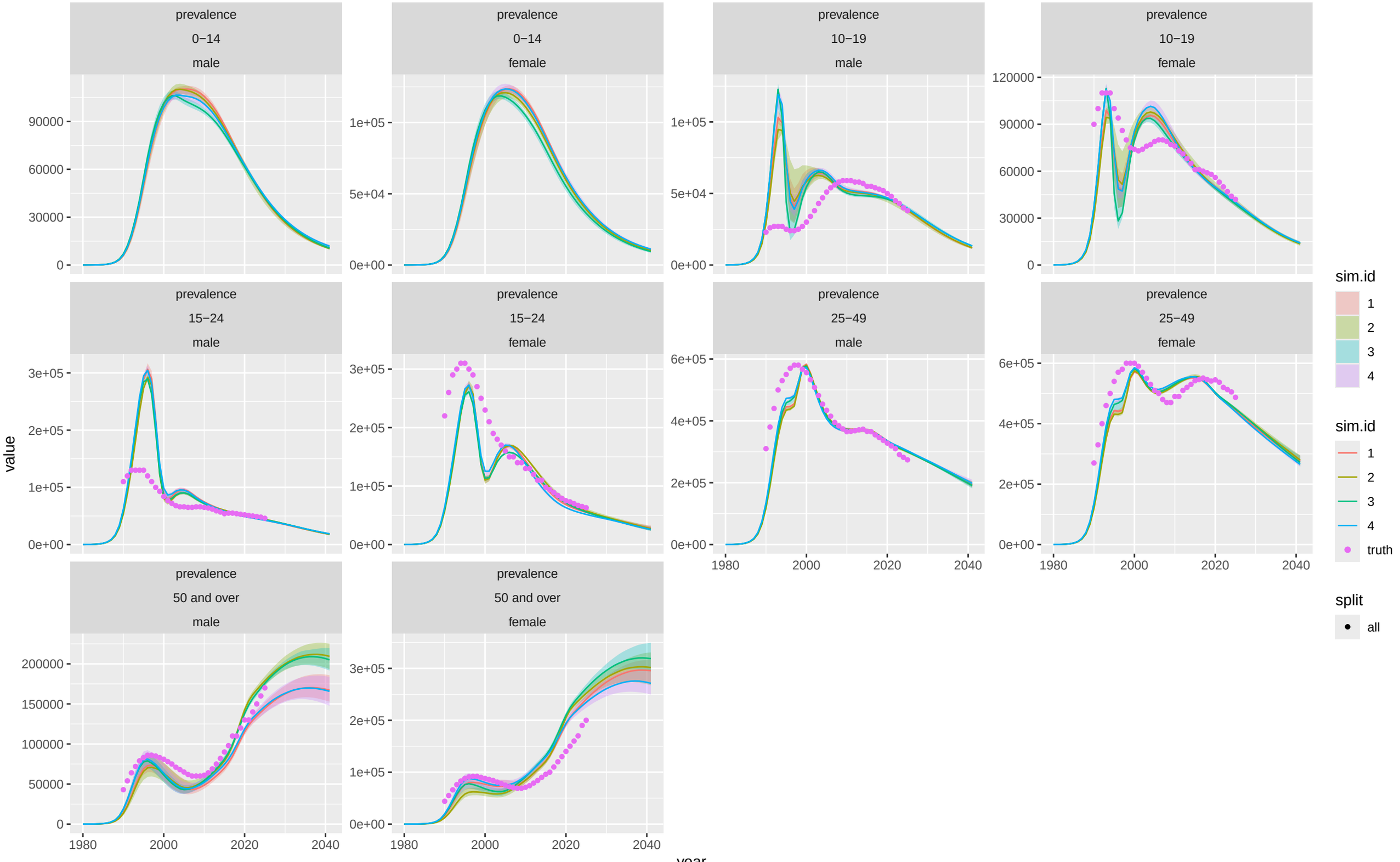
Zimbabwe step 2: hiv.mortality.by.sex (one color per chain)



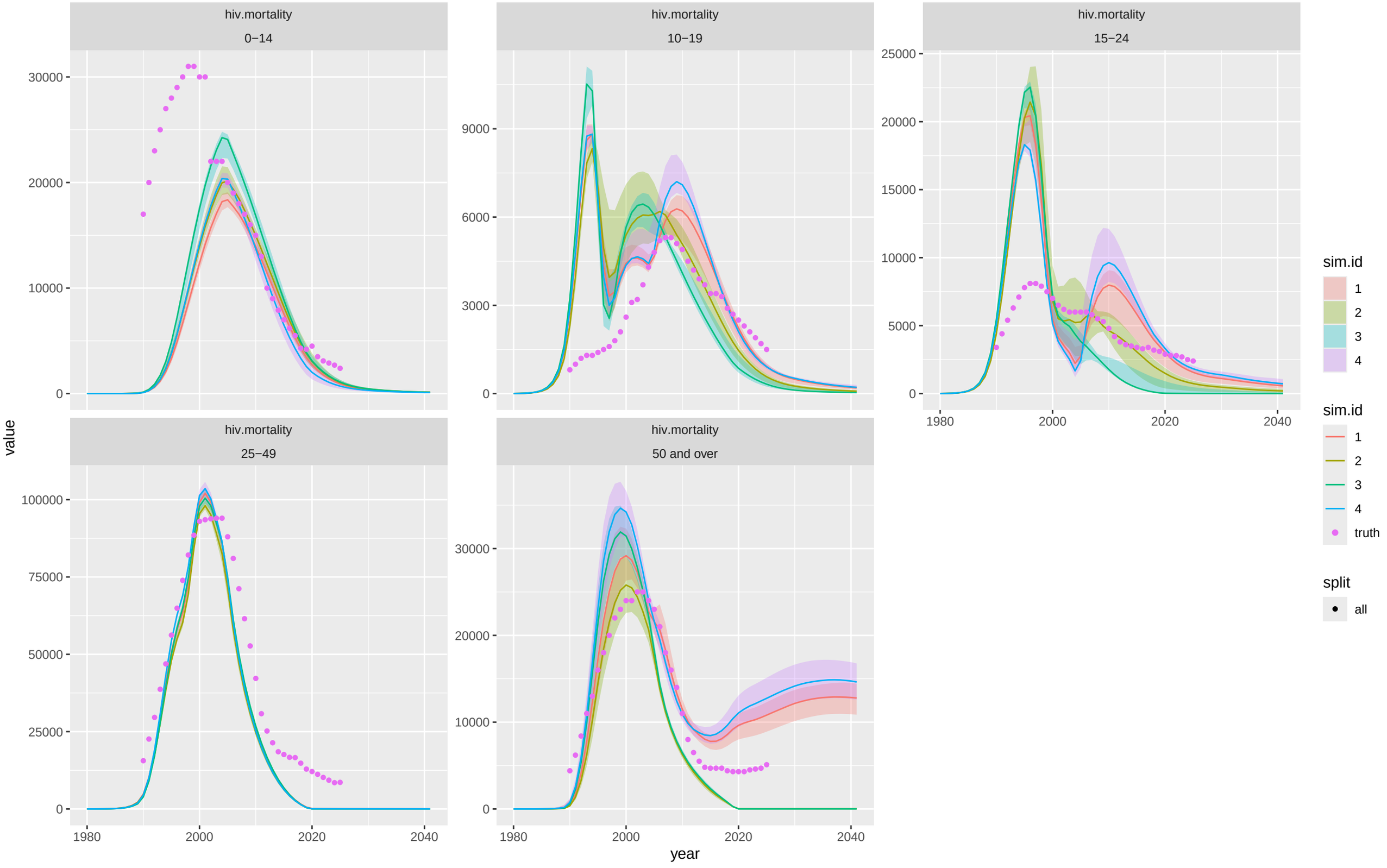
Zimbabwe step 2: incidence.by.age.sex (one color per chain)



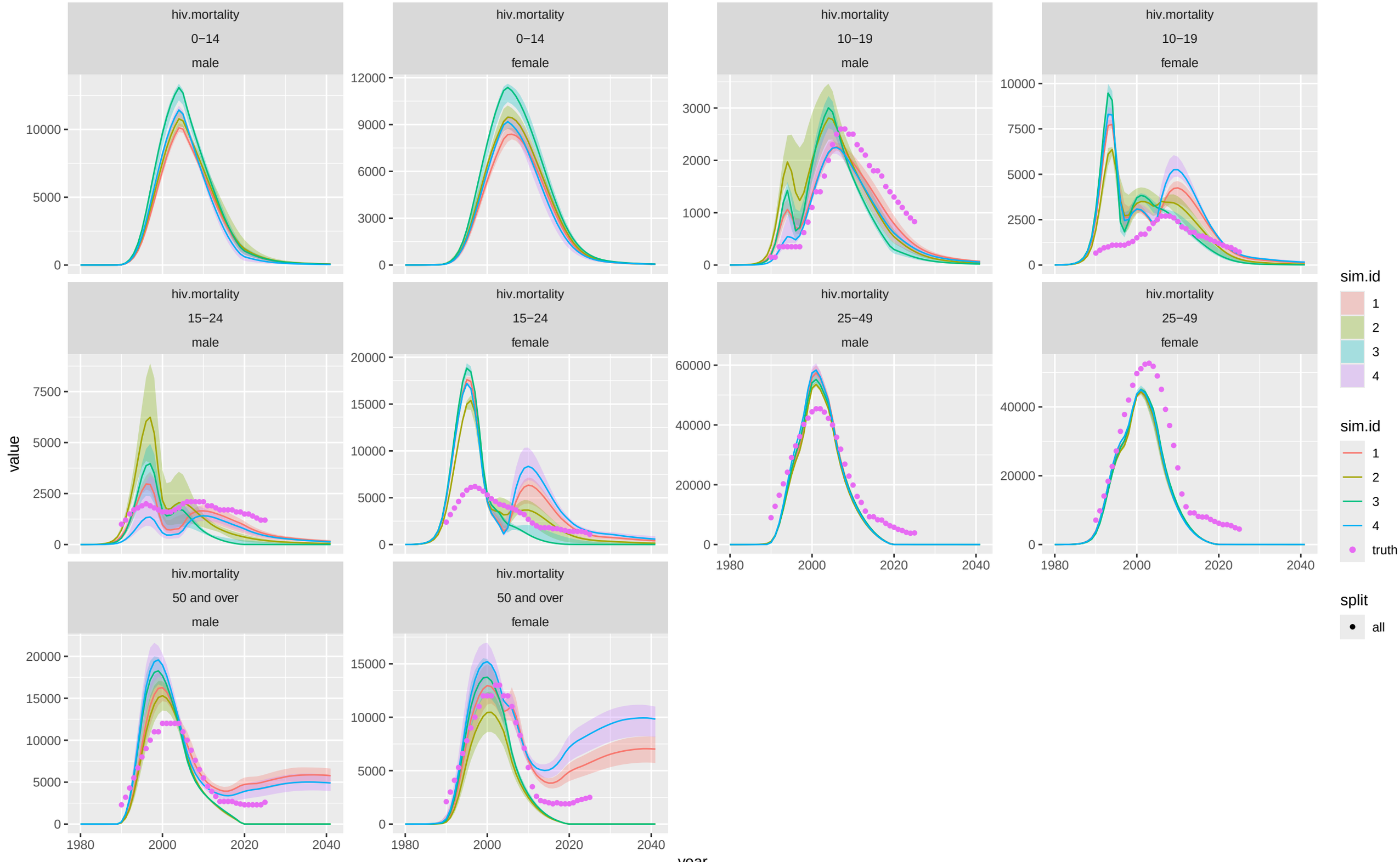
Zimbabwe step 2: prevalence.by.age.sex (one color per chain)



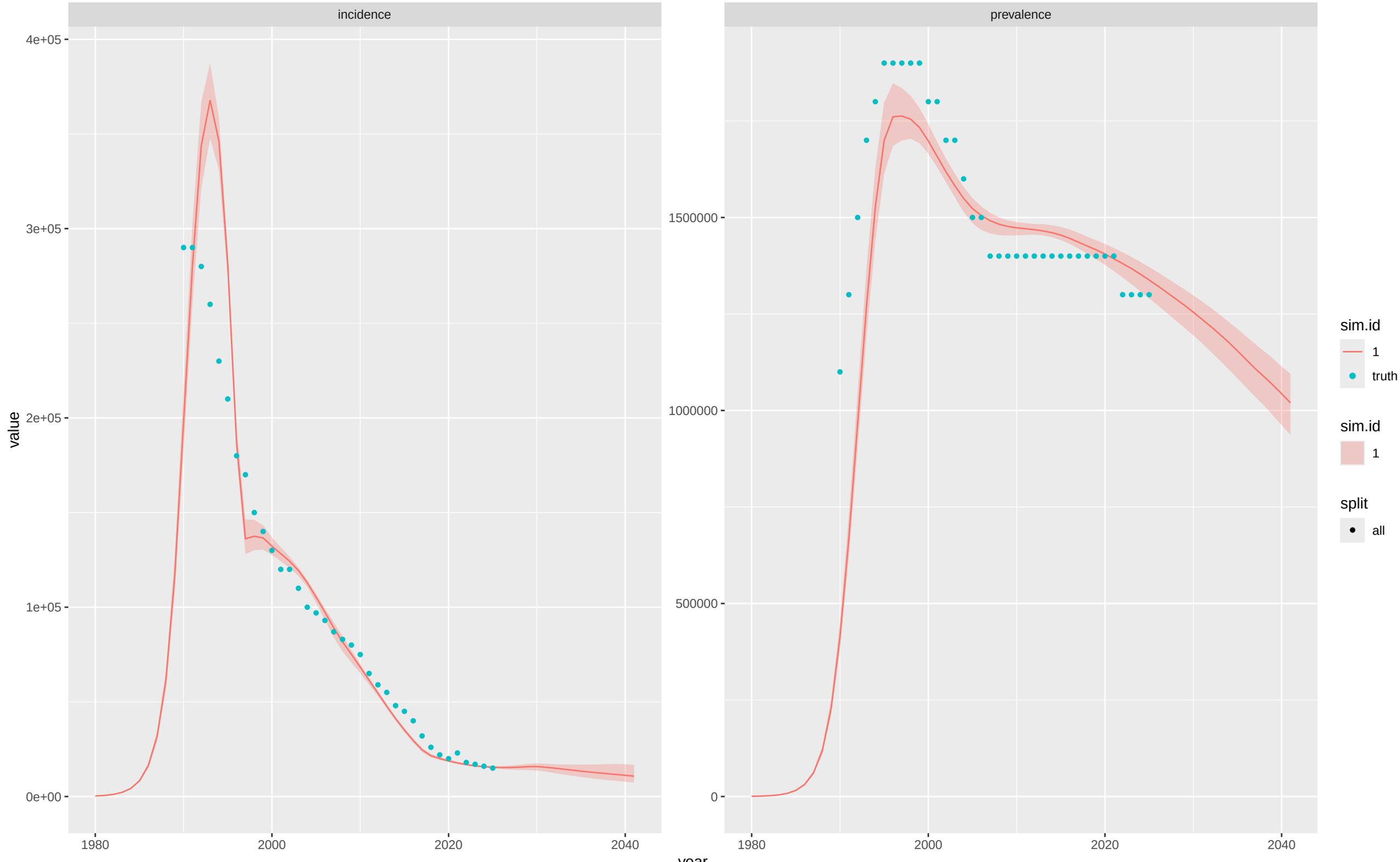
Zimbabwe step 2: hiv.mortality.by.age (one color per chain)



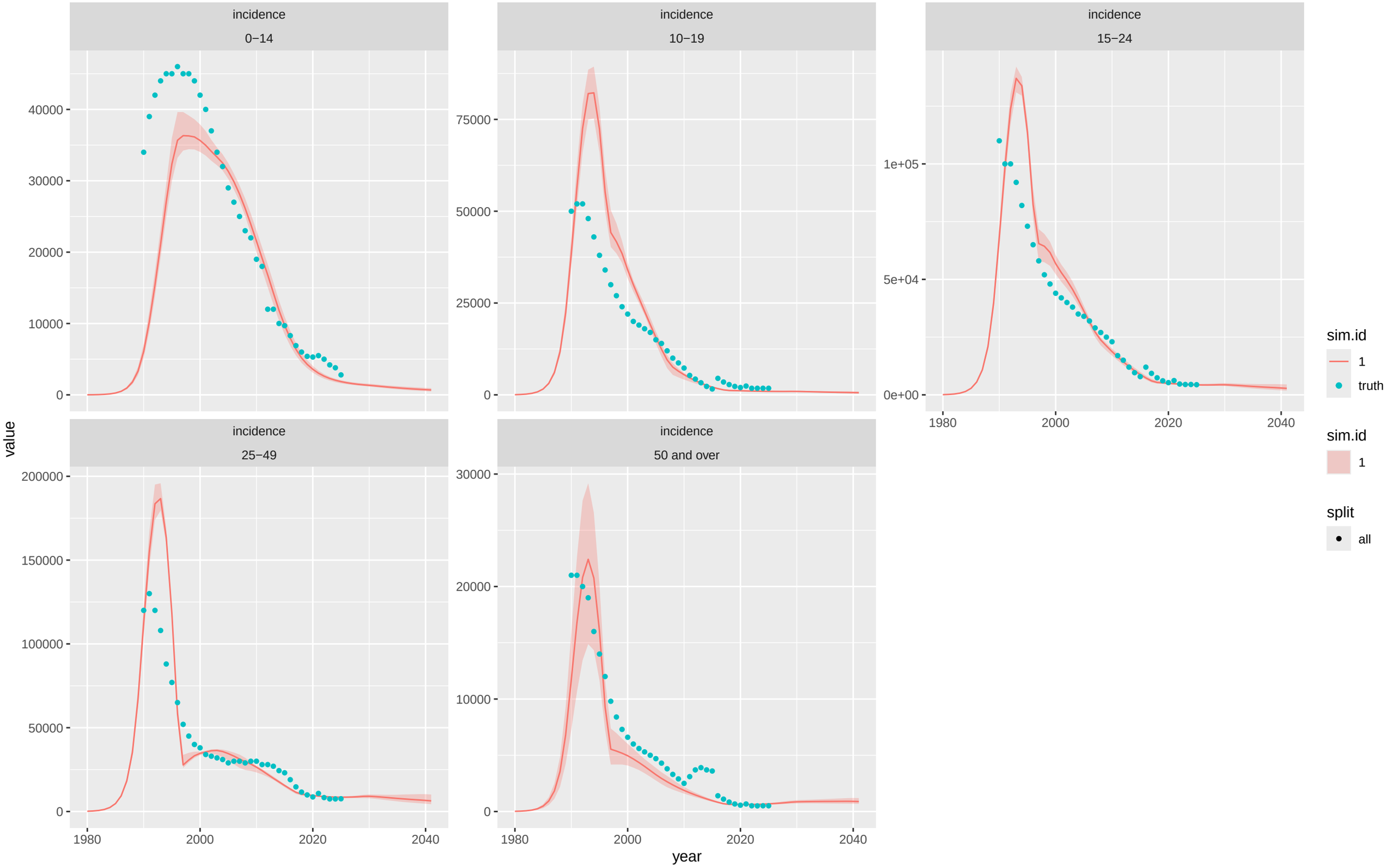
Zimbabwe step 2: hiv.mortality.by.age.sex (one color per chain)



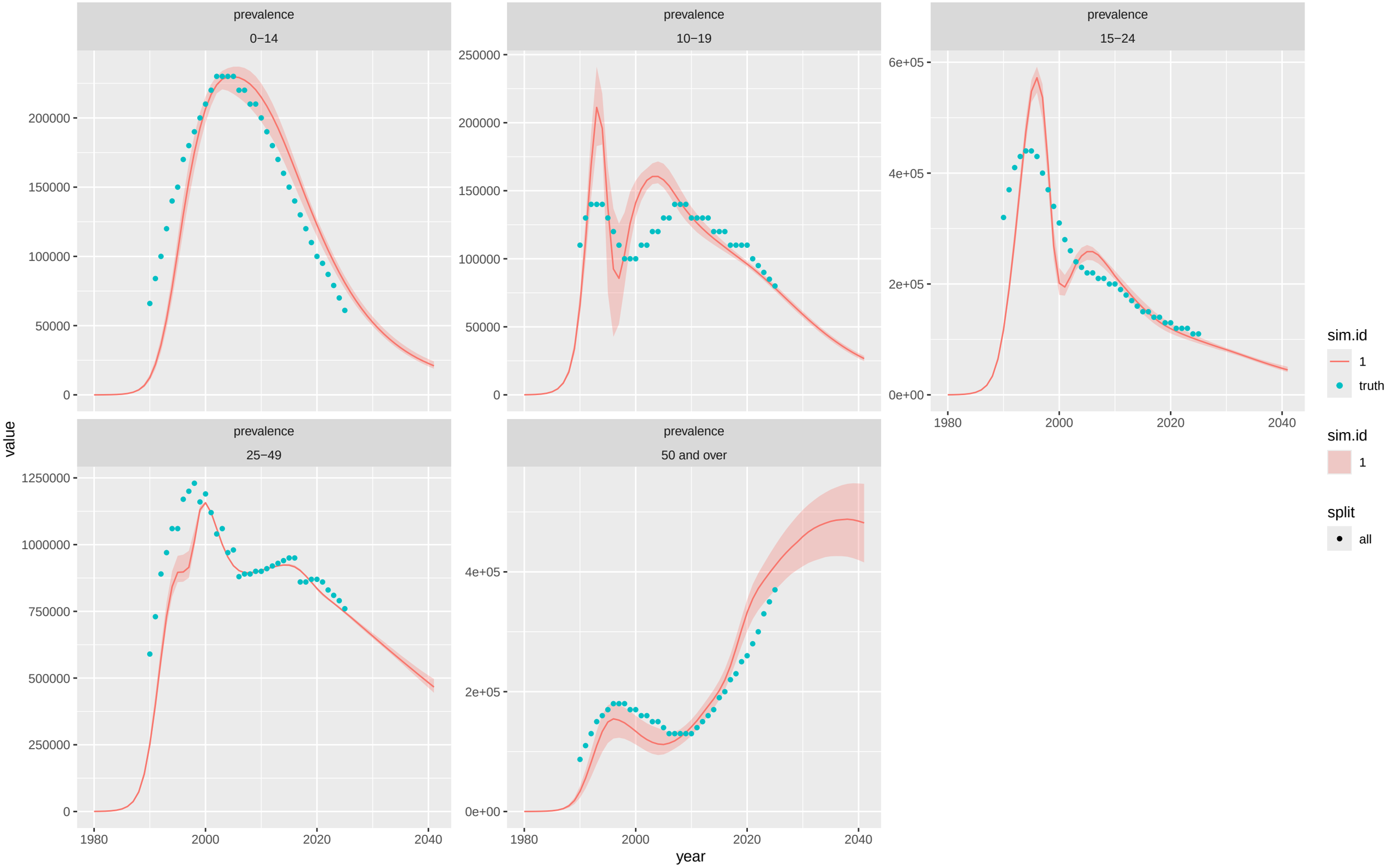
Zimbabwe: fit (merged Step 2 simset)



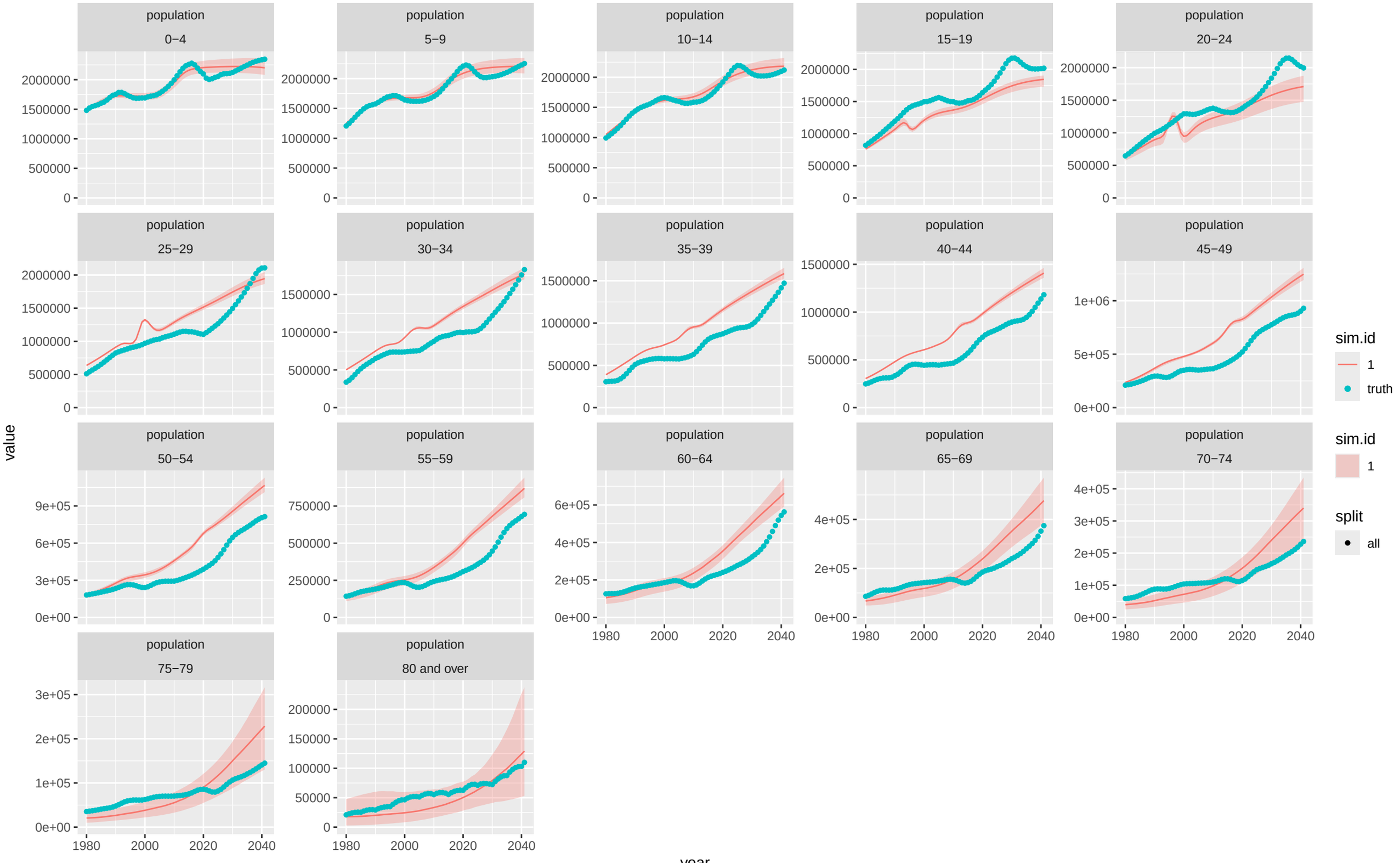
Zimbabwe: incidence.by.age (merged Step 2 simset)



Zimbabwe: prevalence.by.age (merged Step 2 simset)



Zimbabwe: population.by.age (merged Step 2 simset)



Zimbabwe: cascade (merged Step 2 simset)

awareness | engagement | suppression

value

1.00
0.75
0.50
0.25
0.00

1980
2000
2020
2040

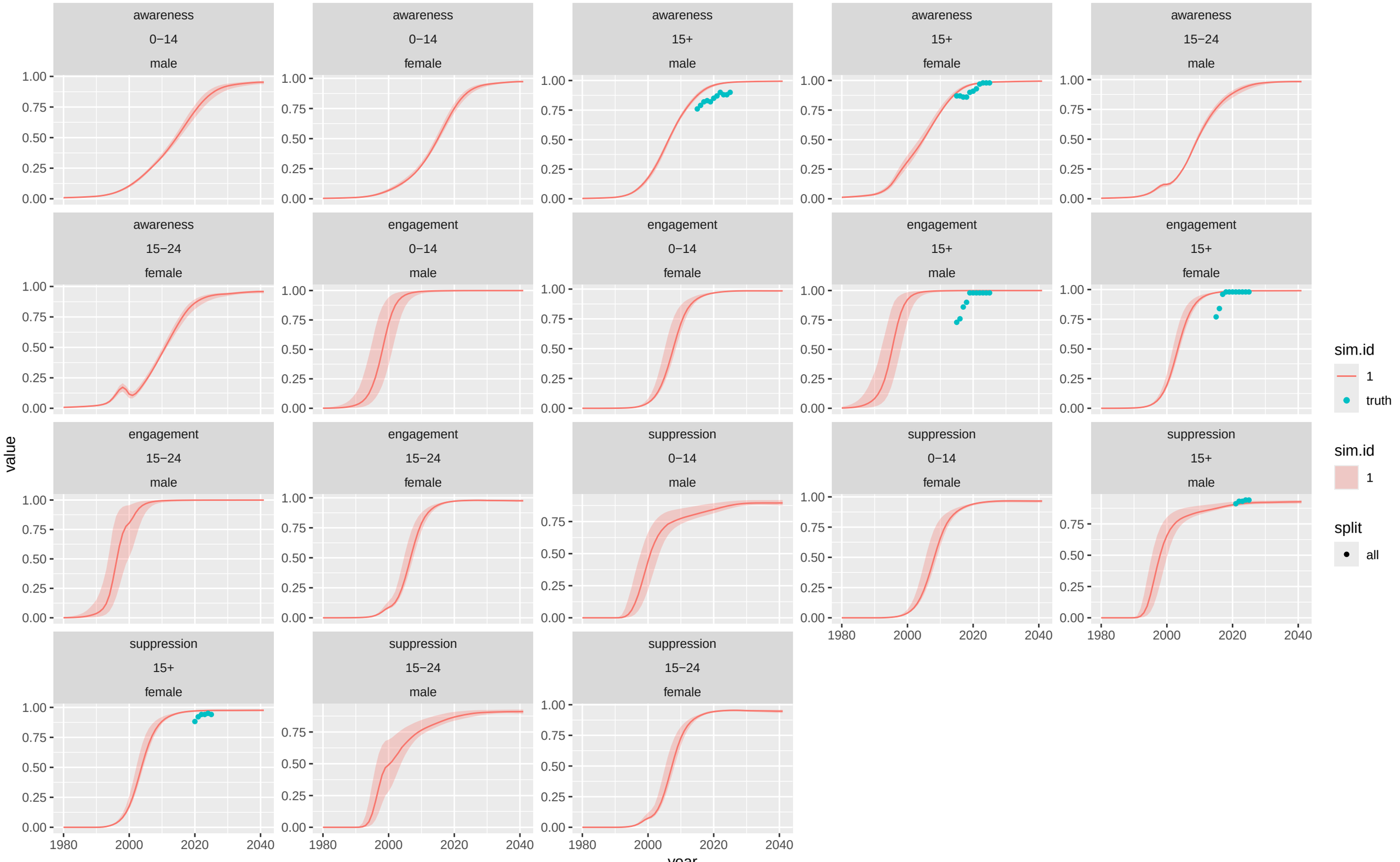
year

sim.id
1
truth

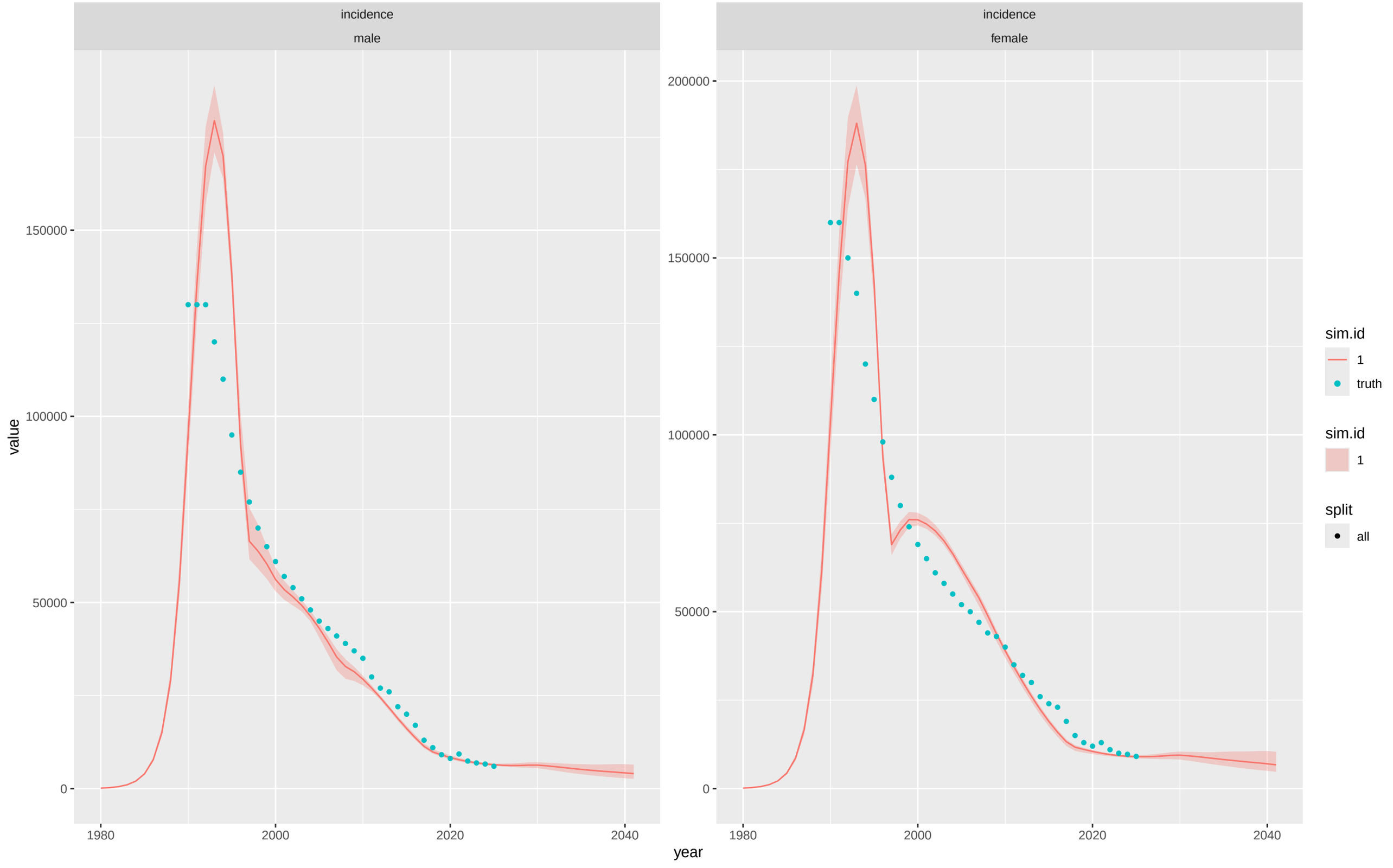
sim.id
1

split
all

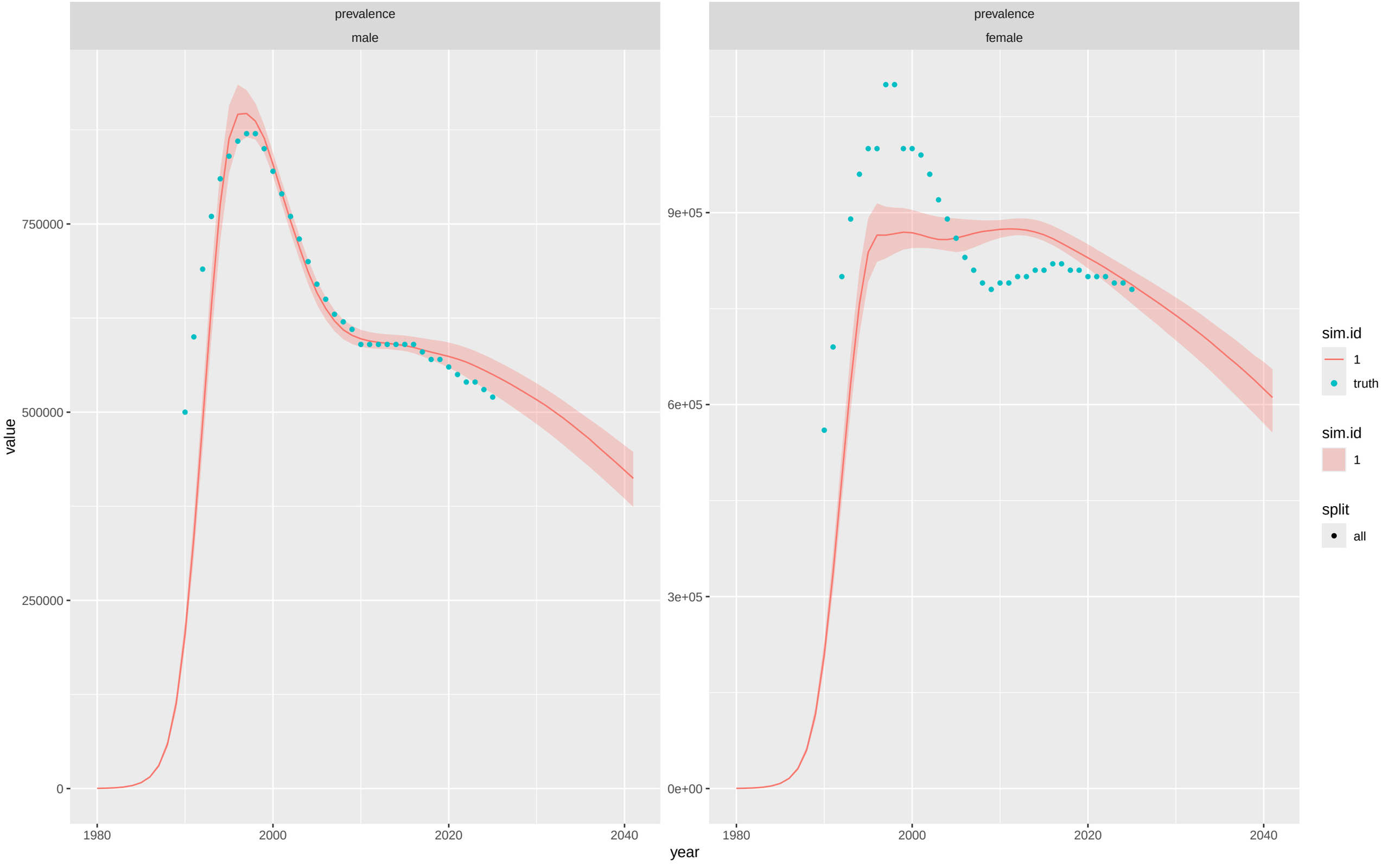
Zimbabwe: cascade.by.age.sex (merged Step 2 simset)



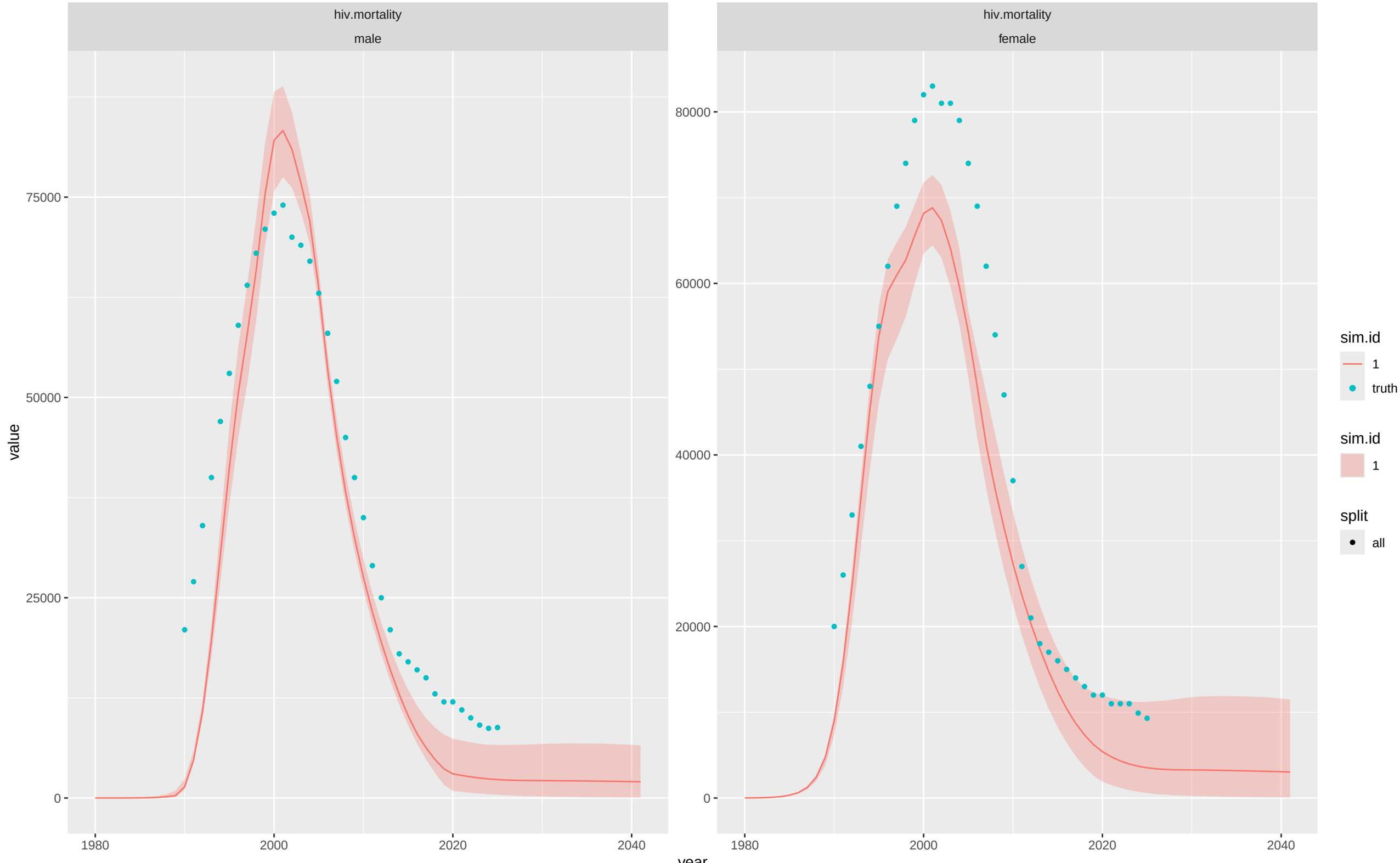
Zimbabwe: incidence.by.sex (merged Step 2 simset)



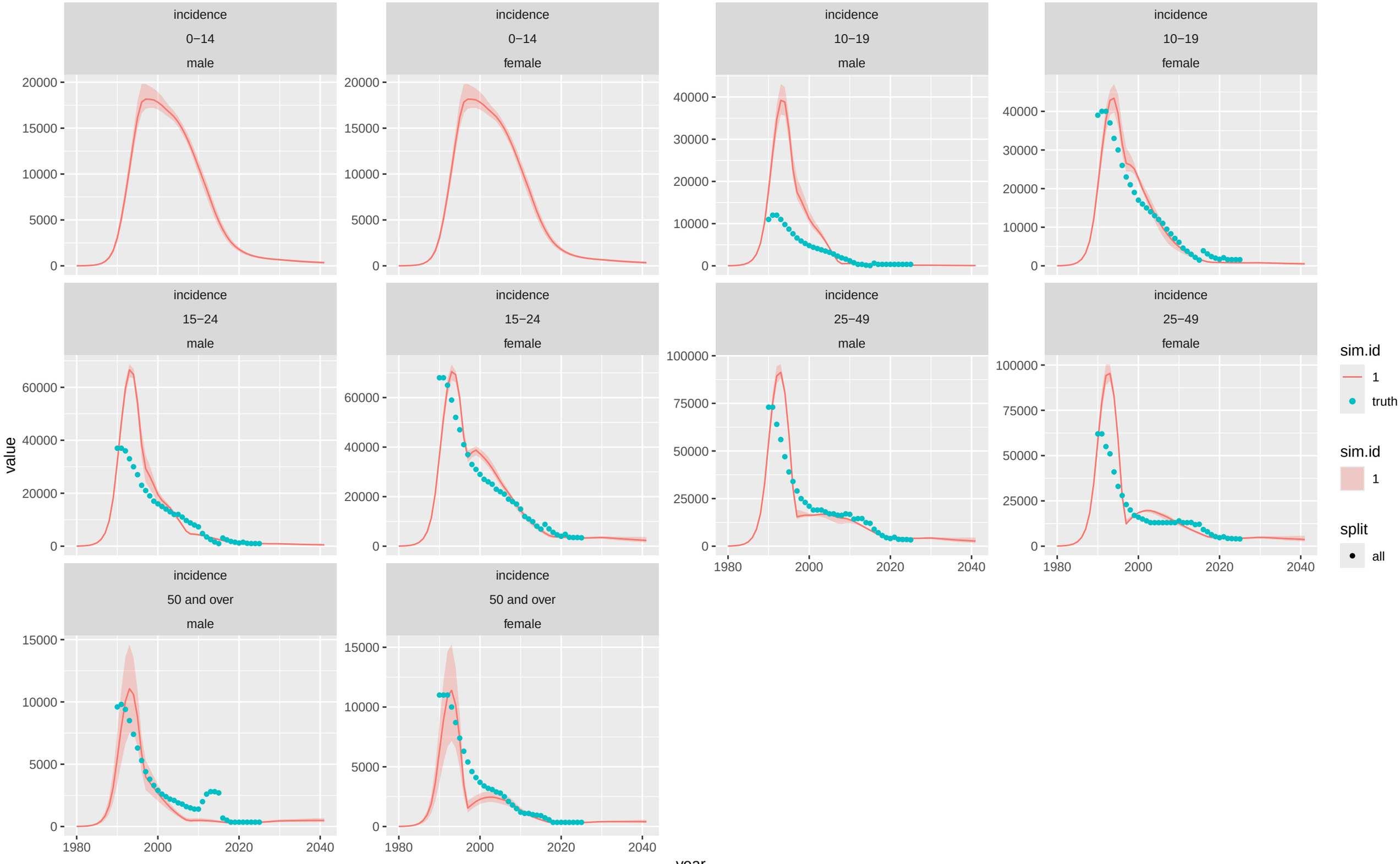
Zimbabwe: prevalence.by.sex (merged Step 2 simset)



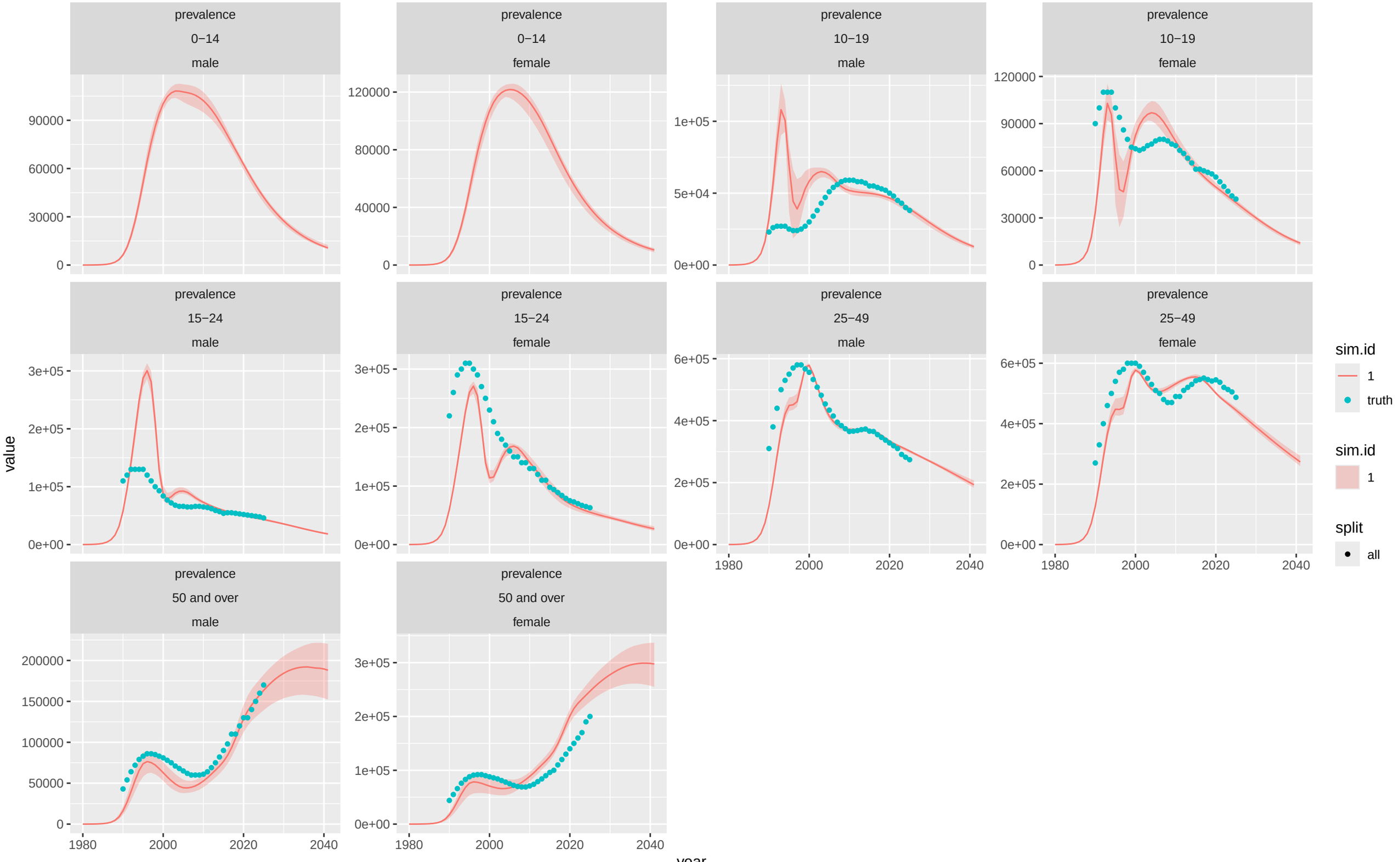
Zimbabwe: hiv.mortality.by.sex (merged Step 2 simset)



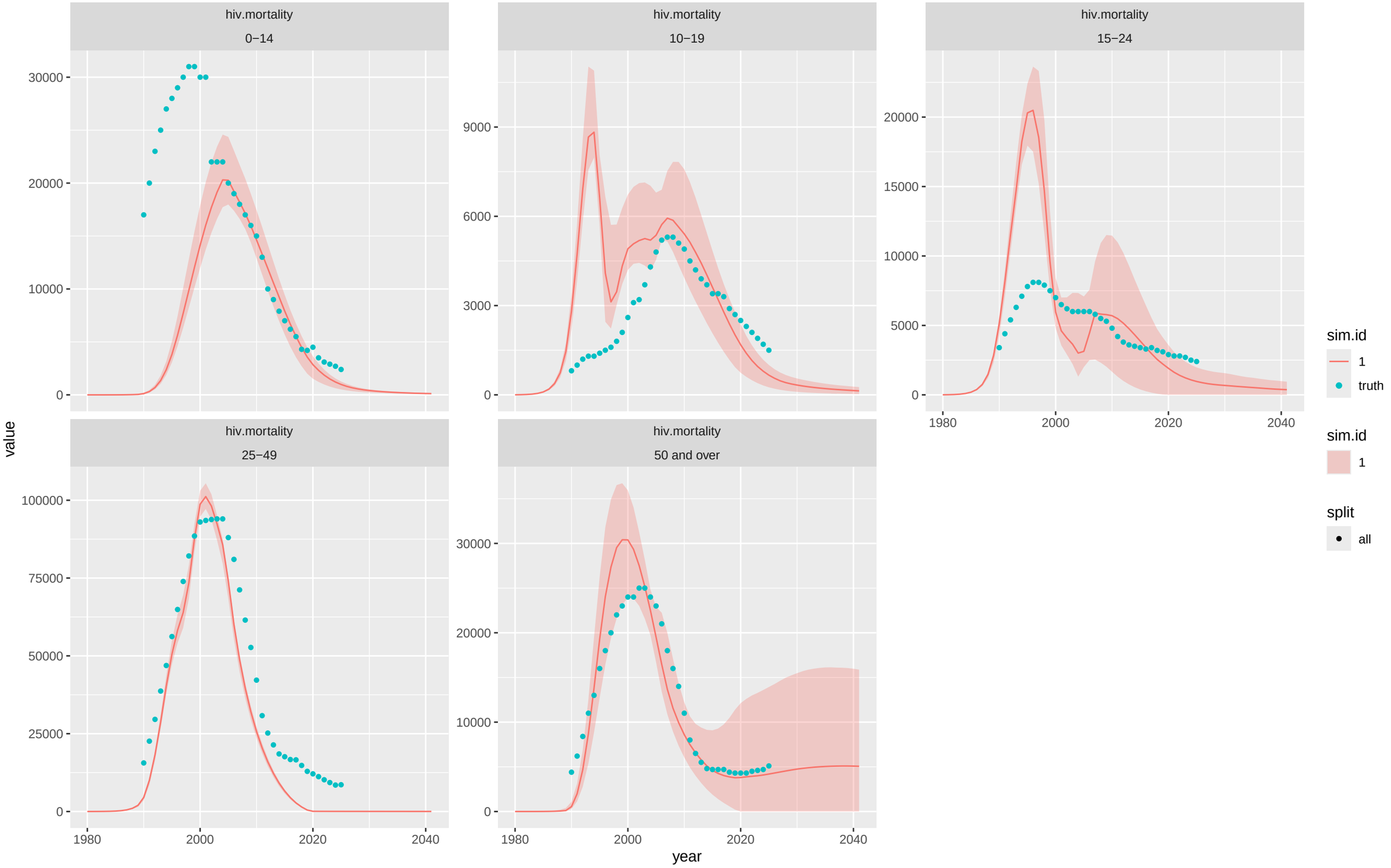
Zimbabwe: incidence.by.age.sex (merged Step 2 simset)



Zimbabwe: prevalence.by.age.sex (merged Step 2 simset)



Zimbabwe: hiv.mortality.by.age (merged Step 2 simset)



Zimbabwe: hiv.mortality.by.age.sex (merged Step 2 simset)

